哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

电子与信息工程学院

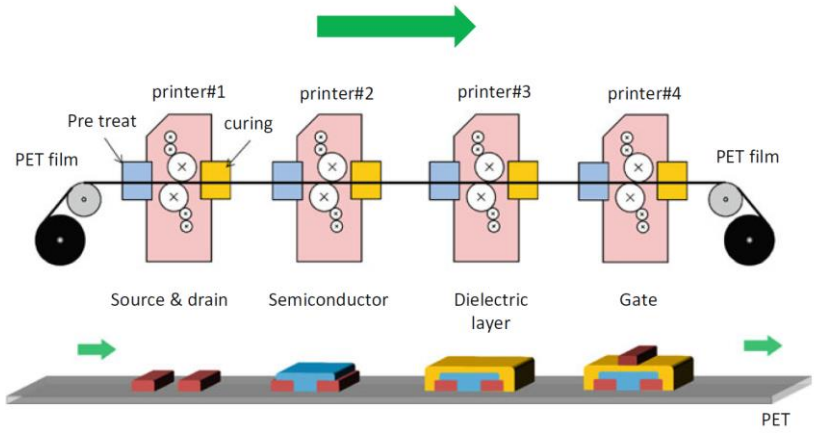
A Lightweight Authentication Scheme with PE-Based Optical Unclonable Label

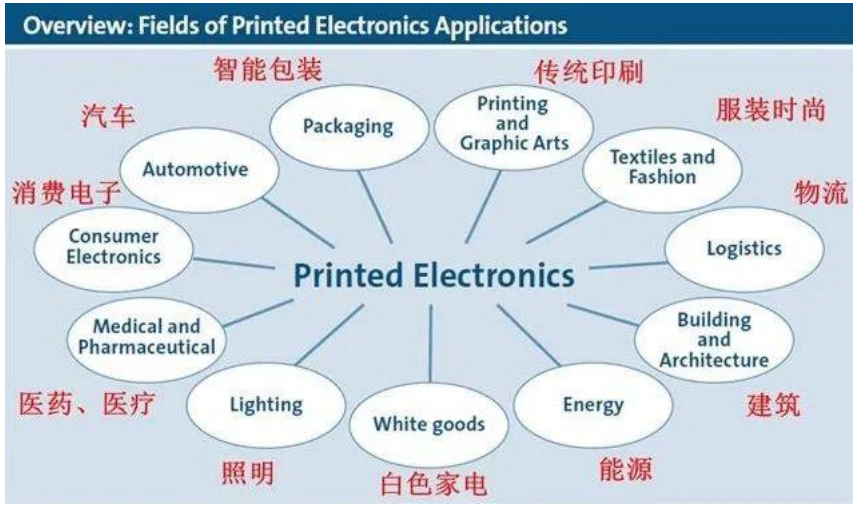
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†Department of Electrical and Computer Engineering, University of Maryland College Park, USA

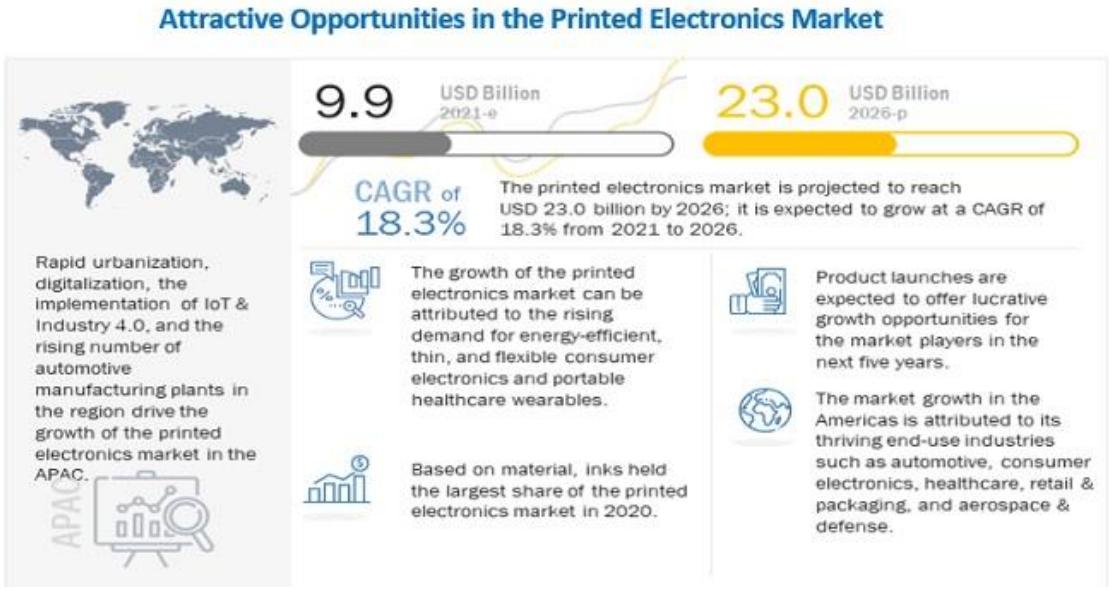
1 Introduction





Advantage:

- Large-scale
- Low-cost
- Customizable
- Environmentally friendly
- Mechanism flexible



How to prevent and identify the counterfeiting of PE products?

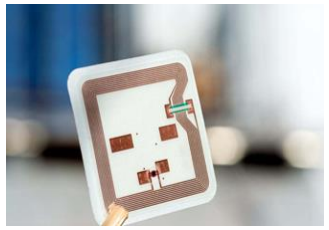
- [1] 2022 white book of flexible electronics industry.
- [2] Printed Electronics Market Size Global Forecast to 2026. https://www.marketsandmarkets.com/Market-Reports/printed-electronics-market-197.html. 2021.

1 Introduction

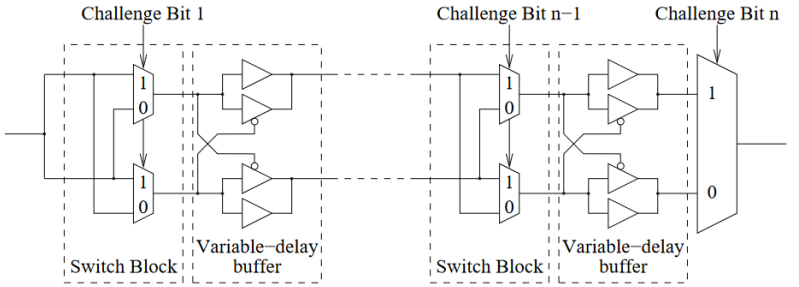
Existing anti-counterfeiting methods

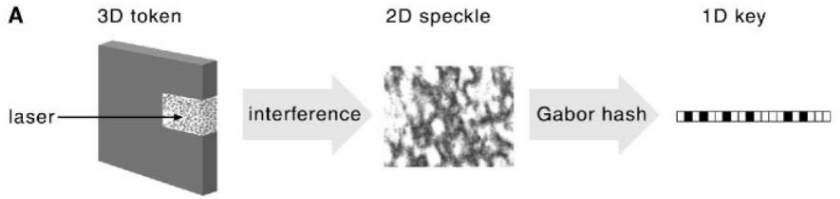






Physical Unclonable Function (PUF)





Disadvantage:
- Vulnerable
- Time-consuming
- High-overhead

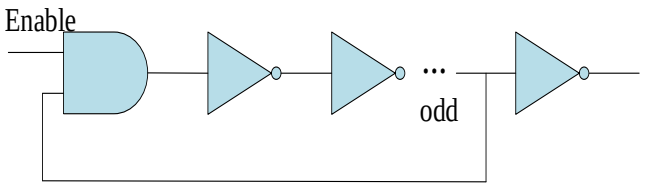
Advantage:
- Unclonable
- Unpredictability
- Uniqueness

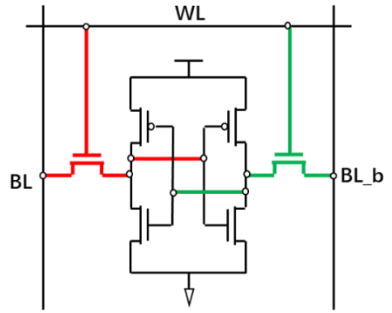
Application:
- IP protection
- Key Generation
- System Authentication

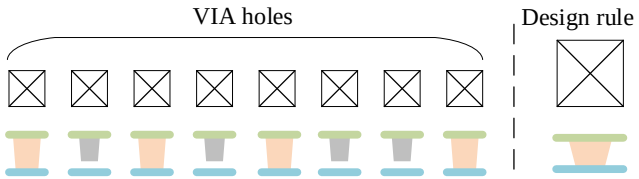
➢ [3] R. Pappu, B. Recht, J. Taylor, and N. Gershenfeld, “Physical One-Way Functions,” Science, vol. 297, no. 5589, pp. 2026, 2002.
➢ [4] B. Gassend, D. Clarke, M. van Dijk, and S. Devadas, ‘‘Silicon physical random functions,’’ in Proc. 9th ACM Conf. Comput. Commun. Security (CCS), 2002, pp. 148–160.

2 **Related Works**: Electronic PUF

- Electronic PUF
  - time domain PUF
    - Ring oscillators PUF [5]
    - Arbiter PUF [6]
  - memory cell PUF
    - SRAM PUF [7]
    - Memristor PUF [8]
  - direct-characterization PUF
    - TV PUF [9]
    - VIA PUF [10]

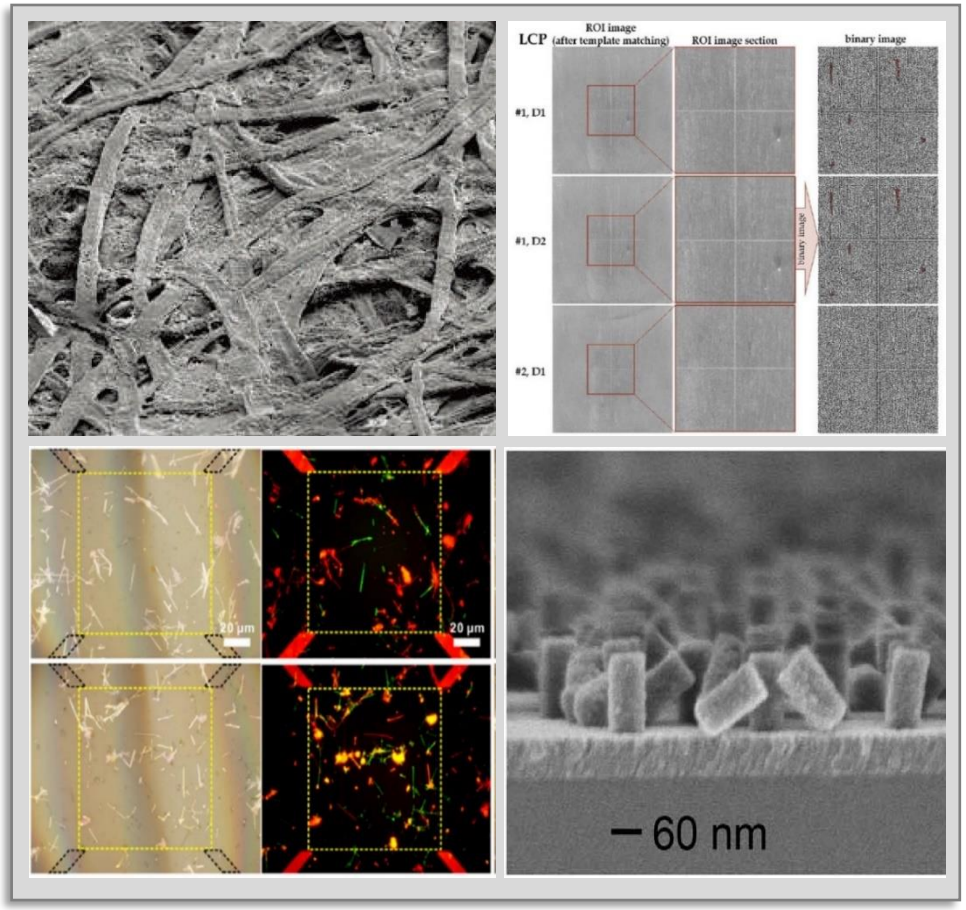






- [5] BOSSUET L, NGO X T, CHERIF Z, et al. A PUF Based on a Transient Effect Ring Oscillator and Insensitive to Locking Phenomenon [J]. IEEE Transactions on Emerging Topics in Computing, 2014, 2(1): 30-6.
- [6] LEE J W, LIM D, GASSEND B, et al. A Technique to Build a Secret Key in Integrated Circuits for Identification and Authentication Applications [C]// 2004 Symposium on VLSI Circuits Digest of Technical Papers (IEEE Cat No 04CH37525), IEEE, 2004.
- [7] GUAJARDO J, KUMAR S S, SCHRIJEN G-J, et al. FPGA Intrinsic PUFs and Their Use for Ip Protection [C]// International workshop on cryptographic hardware and embedded systems, Springer, 2007.
- [8] KOEBERL P, KOCABAŞ Ü, SADEGHI A-R. Memristor PUFs: A New Generation of Memory-Based Physically Unclonable Functions [C]// 2013 Design, Automation & Test in Europe Conference & Exhibition (DATE), IEEE, 2013.
- [9] LOFSTROM K, DAASCH W R, TAYLOR D. IC Identification Circuit Using Device Mismatch [C]// 2000 IEEE International Solid-State Circuits Conference Digest of Technical Papers (Cat No 00CH37056), IEEE, 2000.
- [10] JEON D, BAEK J H, KIM D K, et al. Towards Zero Bit-Error-Rate Physical Unclonable Function: Mismatch-Based Vs. Physical-Based Approaches in Standard Cmos Technology [C]// 2015 Euromicro Conference on Digital System Design, IEEE, 2015.

2 **Related Works**: Optical PUF



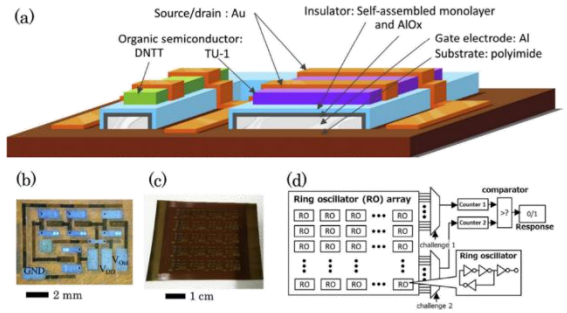
- Paper [11]
- injection moulded plastic [12]
- Nanowires on PET film [13]
- Nano-artifact metrics [14]

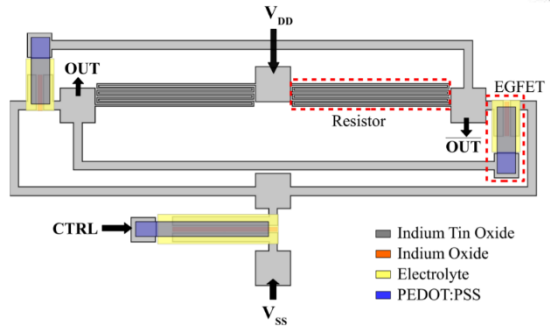
Advantage:
- Lightweight
- Low cost
- Anti-counterfeiting

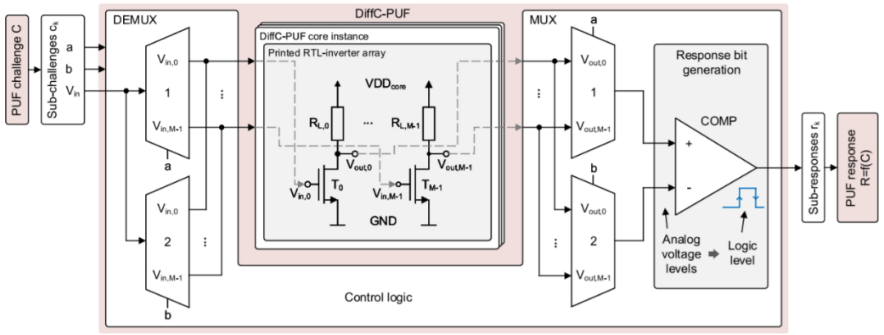
- [11] BUCHANAN J D, COWBURN R P, JAUSOVEC A-V, et al. 'Fingerprinting'documents and Packaging [J]. Nature, 2005, 436(7050): 475.
- [12] WIGGER B, MEISSNER T, FORSTE A, et al. Using Unique Surface Patterns of Injection Moulded Plastic Components as an Image Based Physical Unclonable Function for Secure Component Identification [J]. Scientific reports, 2018, 8(1): 4738.
- [13] Kim J, Yun J M, Jung J, et al. Anti-Counterfeit Nanoscale Fingerprints Based on Randomly Distributed Nanowires [J]. Nanotechnology, 2014, 25(15): 155303.
- [14] MATSUMOTO T, HOGA M, OHYAGI Y, et al. Nano-artifact metrics based on random collapse of resist[J]. Scientific reports, 2014, 4(1): 1-5.

2 **Related Works**: PE-based Electronic PUF

PE PUF
- Printed RO PUF [15]
- Printed Memory PUF [16]
- Diffc-PUF [17]







Disadvantage：
- Complex design
- High cost
- Prone to yield problems

- [15] EROZAN A T, MARQUES G C, GOLANBARI M S, et al. Inkjet-Printed EGFET-Based Physical Unclonable Function—Design, Evaluation, and Fabrication [J]. IEEE Transactions on Very Large Scale Integration (Vlsi) Systems, 2018, 26(12): 2935-46.A. E. Caldwell. et al. Effective iterative techniques for fingerprinting design IP[J]. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 1999, 23(2): 208 - 215.
- [16] KURIBARA K, HORI Y, KATASHITA T, et al. Organic Physically Unclonable Function on Flexible Substrate Operable at 2 V for Iot/Ioe Security Applications [J]. Organic Electronics, 2017, 51: 137-41.
- [17] ZIMMERMANN L, SCHOLZ A, TAHOORI M B, et al. Design and Evaluation of a Printed Analog-Based Differential Physical Unclonable Function [J]. IEEE Transactions on Very Large Scale Integration (Vlsi) Systems, 2019, 27(11): 2498-510.

Related Works: PE-based Optical PUF

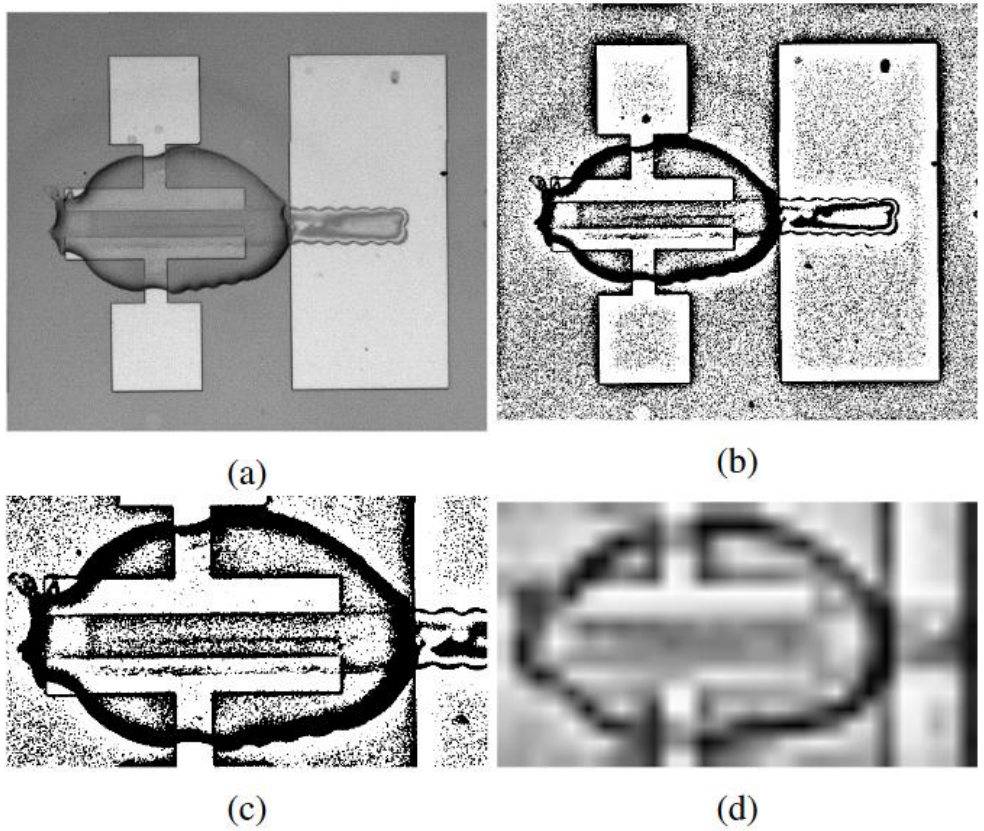


Figure 4: A printed transistor image (a) original, (b) preprocessed, (c) aligned (d) downscaled (downscaling factor=22).

TABLE 1: Definition of datasets

	# of EGTs	Width (μm)	Length (μm)	Image size of ROI after alignment
Dataset-A	18	600	60	350x780
Dataset-B	18	400	60	350x620
Dataset-C	18	200	60	350x460
Dataset-D	18	100	60	350x430

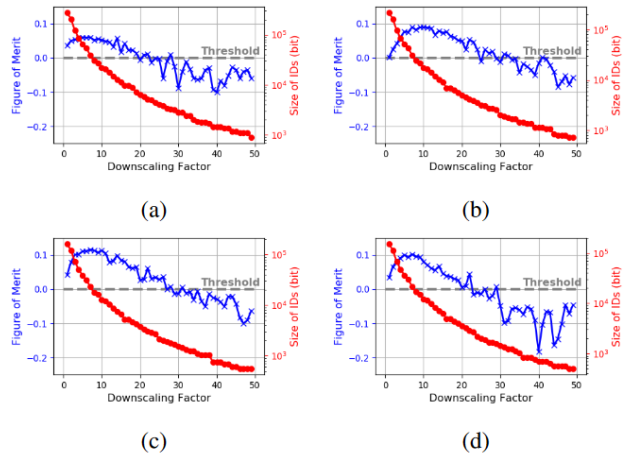


Figure 6: Figure of Merit (FoM) of (a) Dataset-A, (b) Dataset-B, (c) Dataset-C, and (d) Dataset-D with respect to downscaling factor (FoM: difference between minimum value of intra correlation and maximum value of inter correlation).

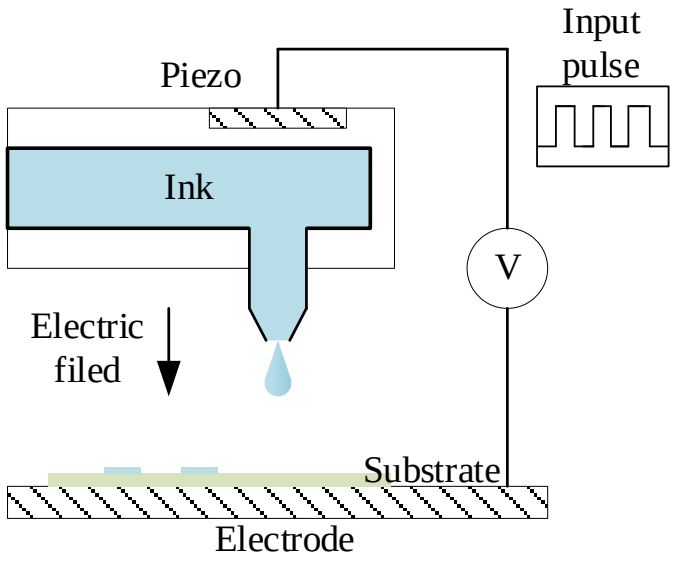
Deficiency:

- Low entropy density
- Require high-precision camera
- Low discrimination

➢ [18] A. T. Erozan, M. Hefenbrock, D. Gnad, M. Beigl, J. Aghassi-Hagmann, and M. Tahoori, "Counterfeit Detection and Prevention in Additive Manufacturing based on Unique Identification of Optical Fingerprints of Printed Structures," IEEE Access, pp. 1–1, 2022.

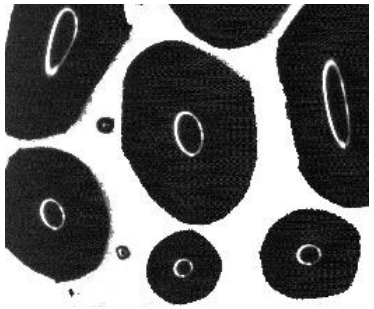
3 **Methodology**: Materials & Parameters

Ink-jet printing



Advantage:

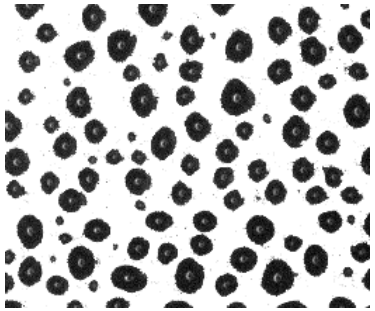
- High accuracy
- No mask
- Flexible



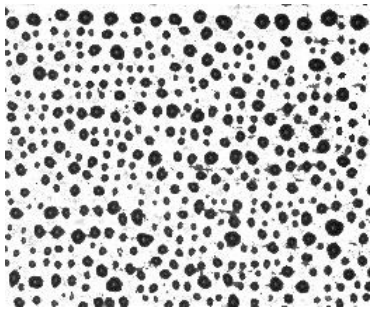
DS=10 μm



DS=20 μm



DS=35 μm



DS=50 μm



PET film



Aqueous nano-silver ink

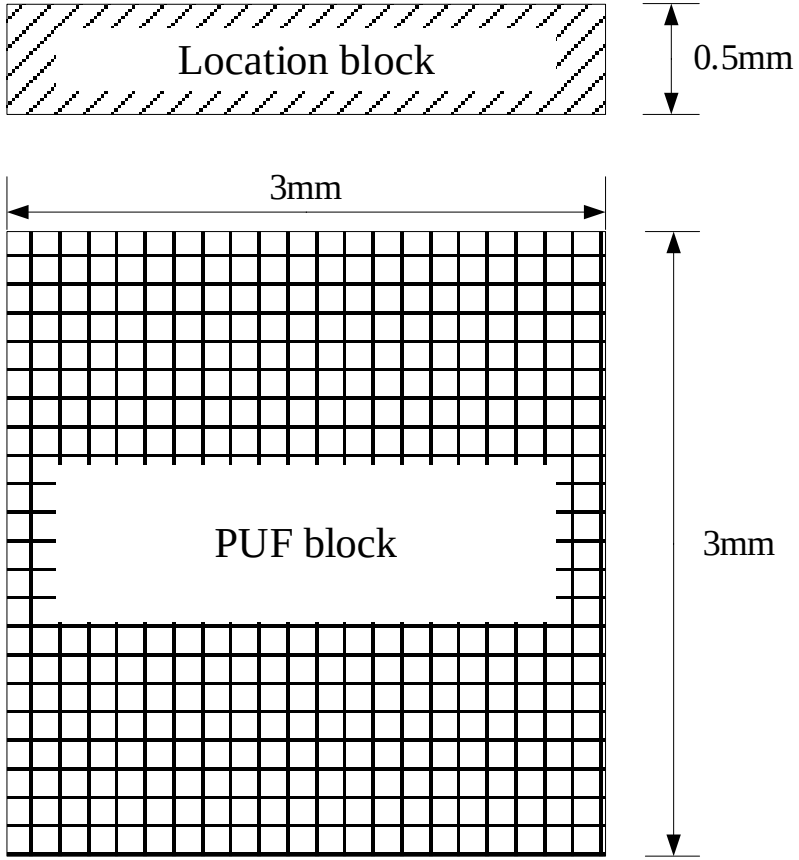
Waveform: slow slew rate1 Drop spacing: 20μm

Print height: 0.45mm Nozzle Voltage: 16V

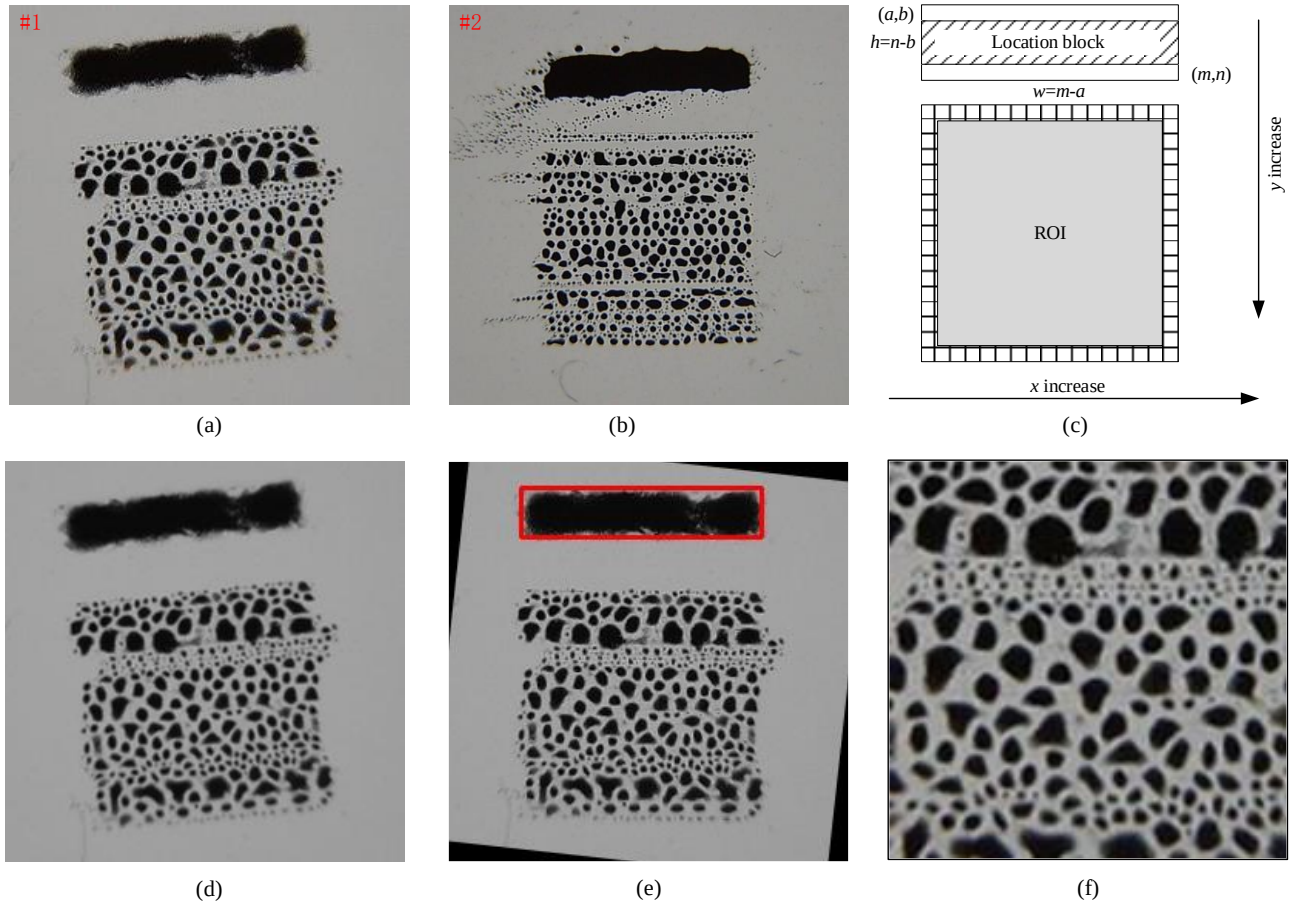
- [19] M. Singh, H. M. Haverinen, P. Dhagat, and G. E. Jabbour, "Inkjet printing—process and its applications," Advanced materials, vol. 22, no. 6, pp. 673-685, 2010.
- [20] Z. Cui, Printed electronics: materials, technologies and applications. John Wiley & Sons, 2016.

3 **Methodology**: Optical Unclonable Label (OUL)

Prototype

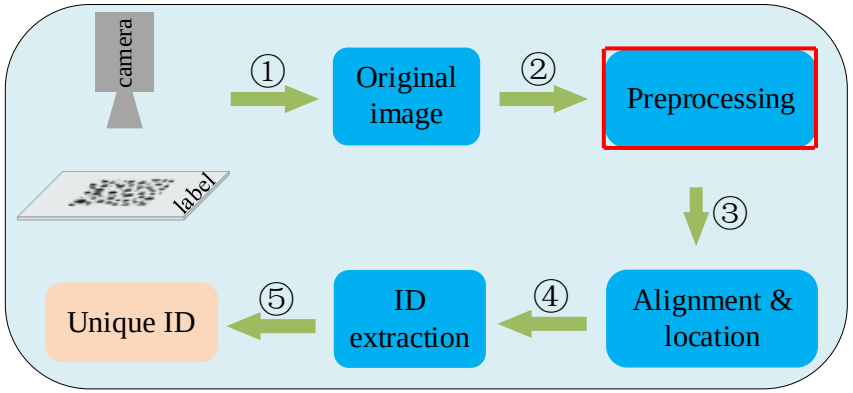


Illustration



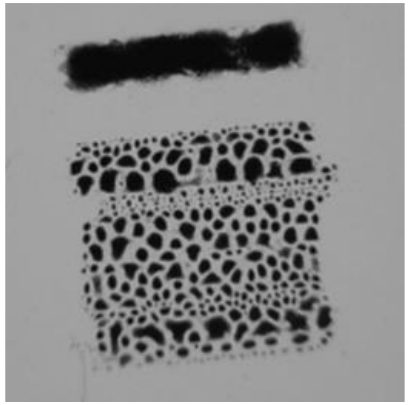
(a) and (b) are original OULs, #1 and #2; (c) is the principle of location; (d) preprocessing; (e) alignment and location; (f) located ROI.

3 **Methodology**: ID Extraction Algorithm





original OUL



after preprocessing

Purpose:

- Reduce noise
- Reduce the influence of light on the results
- Facilitate subsequent processing

Part 1: preprocessing

procedure

Input: Ori.image

Output: Gray.image

1.RBG2Gray (){

$$Gray = R \times 0.299 + G \times 0.587 + B \times 0.114$$

}

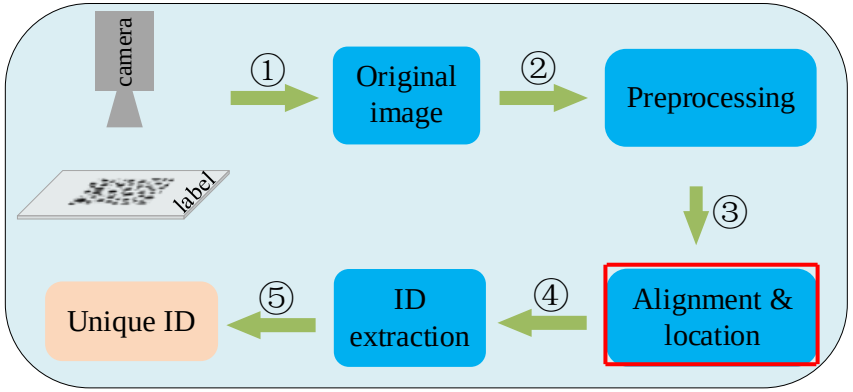
2.Gaussian Filter(){

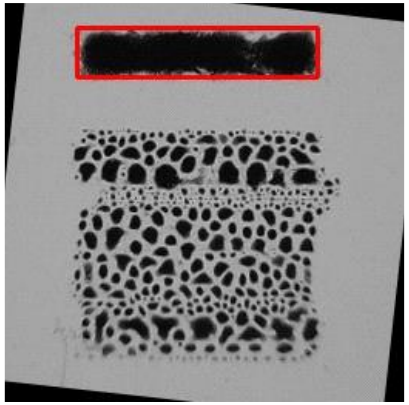
$$new\ pixel = \begin{bmatrix} 0.0585 & 0.0965 & 0.0585 \\ 0.0965 & 0.1529 & 0.0965 \\ 0.0585 & 0.0965 & 0.0585 \end{bmatrix} \times pixel\ window$$

}

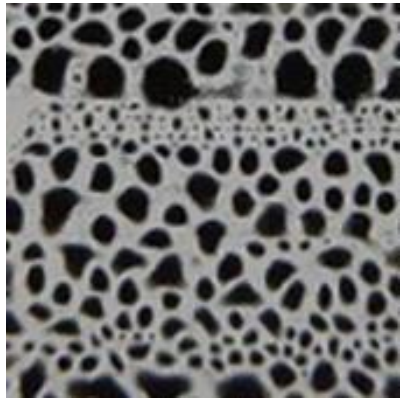
end procedure

3 **Methodology**: ID Extraction Algorithm





alignment



location

Purpose:

- Locate ROI
- Increase information entropy density

Part 2: alignment & location

procedure

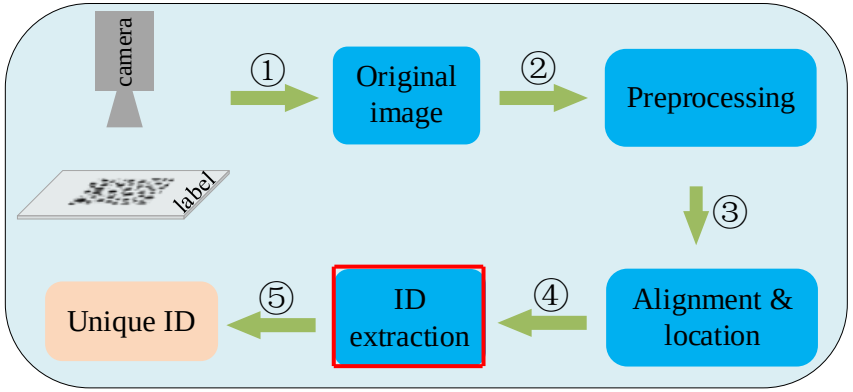
Input: Gray.image

Output: ROI

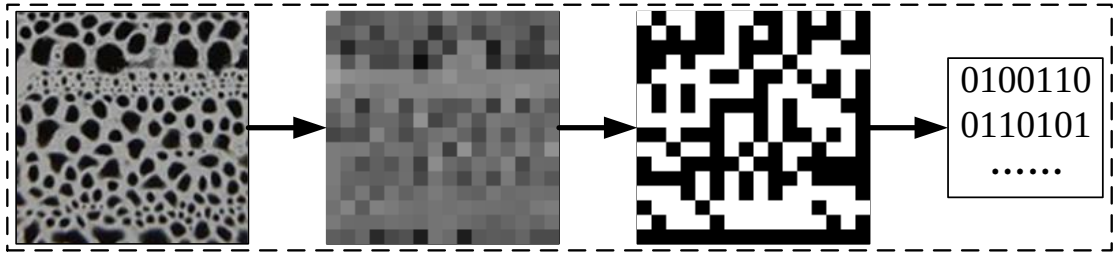
1. Image binarization
2. Location block detection
3. Transformation matrix generation
4. Image alignment
5. ROI location

end procedure

3 **Methodology**: ID Extraction Algorithm



example



Purpose:

- Generate unique ID

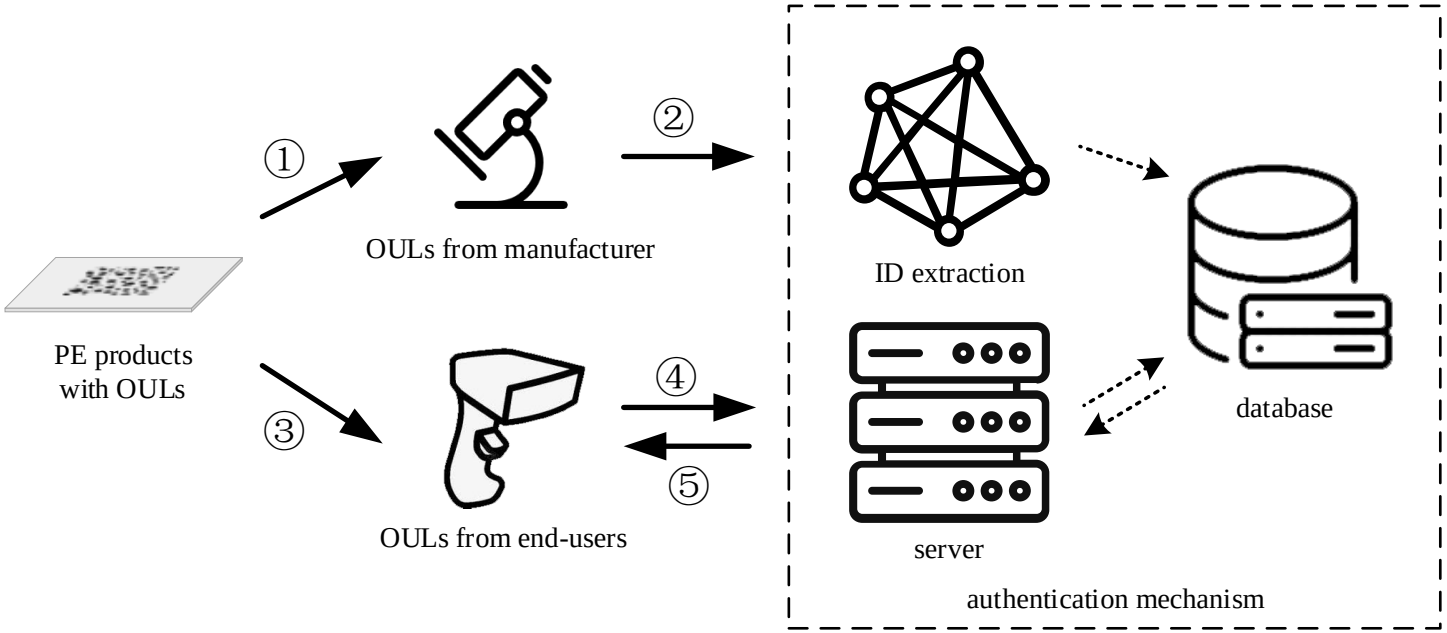
Part 3: ID generation

procedure
Input: ROI
Output: binary ID
1. Reduce size
2. Calculate the average value
3. Compare the grayscale of the pixels and generate responses
4. Sort responses to obtain hash value

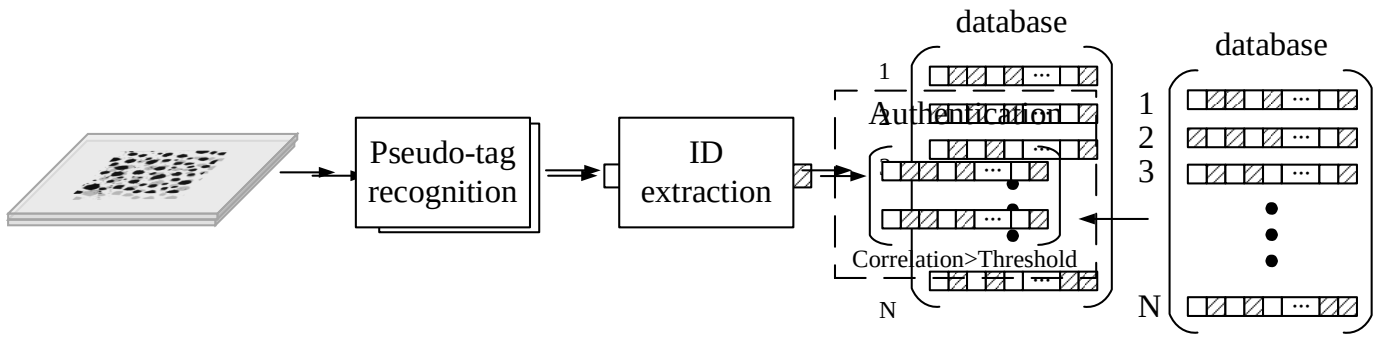
end procedure

3 **Methodology**: Lightweight authentication system

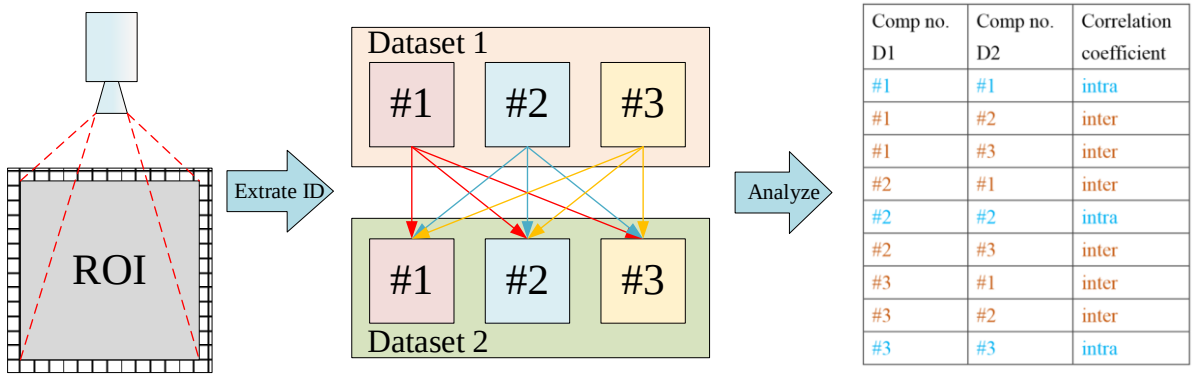
Overview



Validation phase



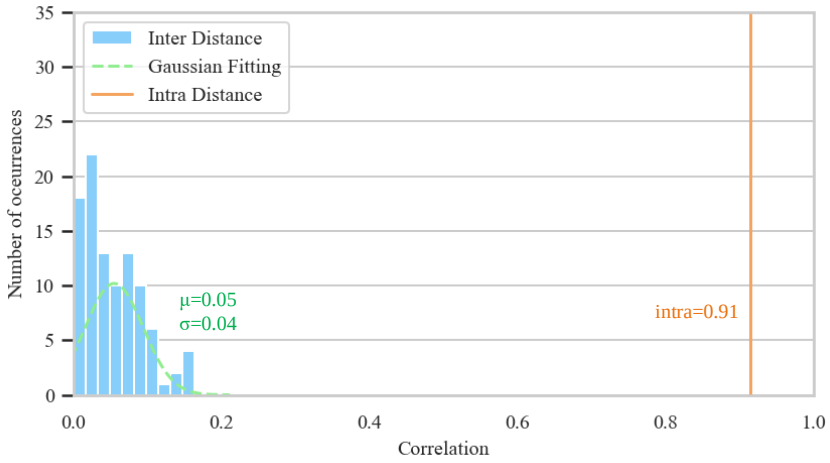
4 Results & Discussion



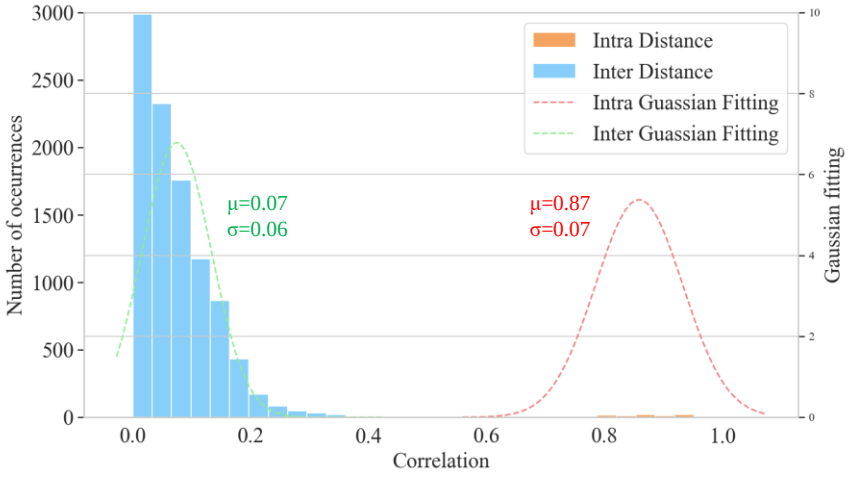
Comp no. D1	Comp no. D2	Correlation coefficient
#1	#1	intra
#1	#2	inter
#1	#3	inter
#2	#1	inter
#2	#2	intra
#2	#3	inter
#3	#1	inter
#3	#2	inter
#3	#3	intra

Identification Certainty:

$$\zeta_n = \frac{\alpha_n - \mu_{\text{inter},n}}{\sigma_{\text{inter},n}}$$



(a) single OUL



(b) all OULs

single OUL:

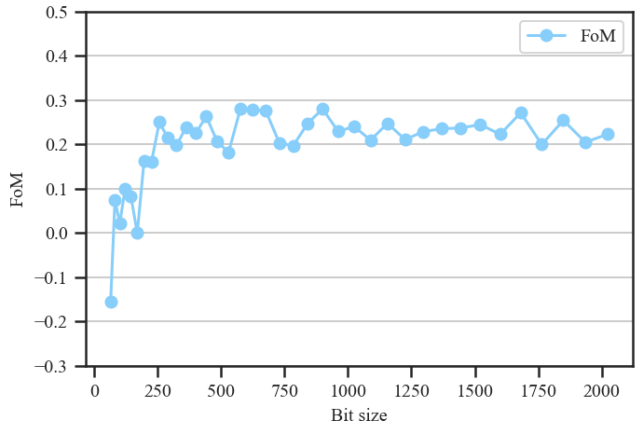
- average ζ_n of 21.5σ.

all OULs:

- ζ_n of 13.3σ.

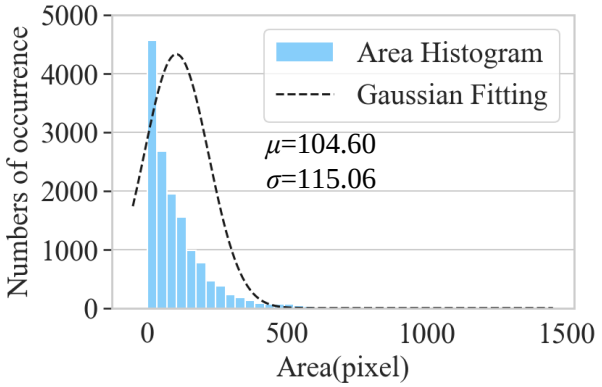
4 Results & Discussion

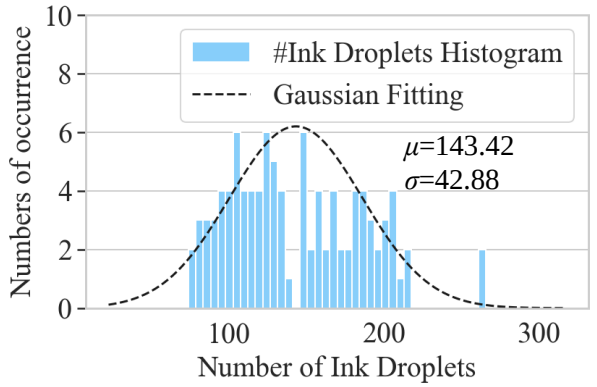
Figure of Merit: $$FoM(I) = \min_{\{(i,j)|i=j,i,j\in I\}} C(i,j) - \max_{\{(i,j)|i\neq j,i,j\in I\}} C(i,j)$$



Optimal FoM: 0.29

Compress to 256bit, FoM is 0.25.





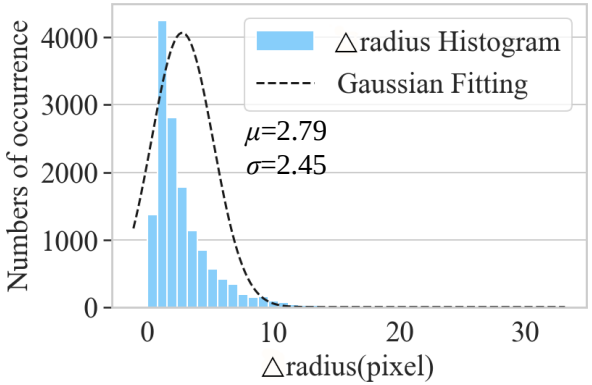


TABLE I. MAX FOM AND REQUIRED BIT SIZE OF DIFFERENT METHODS

Scheme	Bit size	FoM
iPUF [18]	256	< 0
	2400	0.15
Proposed	256	0.25
	600	0.29

TABLE II. DISTRIBUTION OF THREE JUDGMENT PARAMETERS

Parameter	μ	σ	Confidence interval
area	104.60	115.06	[0, 334.72]
$\Delta radius$	2.79	2.45	[0, 7.69]
#ink droplets	143.72	42.88	[57.96, 229.48]

- [18] A. T. Erozan, M. Hefenbrock, D. Gnad, M. Beigl, J. Aghassi-Hagmann, and M. Tahoori, "Counterfeit Detection and Prevention in Additive Manufacturing based on Unique Identification of Optical Fingerprints of Printed Structures," IEEE Access, pp. 1–1, 2022.

5 Conclusion

An Optical Unclonable Label Based on PE Technology

- Fully compatible with PE process and low cost
- The pattern of OUL is flexible enough, releasing from the constrains of the circuit.
- ID extraction algorithm
- Good identification certainty and FoM

A Lightweight Authentication Scheme Based on The Proposed OUL

- Lightweight and low storage cost
- Pseudo tag recognition algorithm to defend machine learning attack
- Good reliability and high security

Future work

- Try more combinations of ink and substrate
- Improved ID extraction algorithm



THANKS!

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